GUJARAT TECHNOLOGICAL UNIVERSITY
Chandkheda, Ahmedabad
Affiliated

Government Engineering College,Gandhinagar
A Report On

GSRTC PASS GENERATION SYSTEM AND ONLINE BUS TRACK

Under subject of DESIGN ENGINEERING II-B
B. E. Semester: VI
(IT Engineering)

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CANDIDATE'S DECLARATION

We hereby declare that the work presented in this project entitled **"GSRTC ONLINE BUS TRACKING AND PASS GENERATION SYSTEM"** submitted towards completion of project in Third Semester of B.E. (IT) is an authentic record of our original work carried out under the guidance of **"Prof. Prashant chaudhary"**. We have not submitted the matter embodied in this project for the award of any other degree.

Semester: 6th

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CERTIFICATE

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This is to certify that the "GSRTC ONLINE BUS TRACKING AND PASS GENERATION SYSTEM has been carried out by PATEL BHUMI, GRACY RATHOD, AYUSHI PRAJAPATI, RINKI DEVI under my guidance in fulfillment of the subject Design Engineering in Information Technology Engineering (6" Semester) of Gujarat Technological University, Ahmadabad during the academic year 2024-25.

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ABSTRACT

Gujarat state faces severe problems of road congestion and associated issues of commuters, which include delays in the arrival of buses at bus stops, lack of information about different bus routes and stops and time. College students staffs will miss their bus by a fraction of second. This will lead to many problems like being late for the classes and sometimes late for the exams also. To overcome this, we have implemented a Smart Bus Tracking System. This system is used to track the GSRTC state transport buses.

The proposed system uses a Smart phone application. Buses carry Global Positioning System (GPS) devices to track their positions and Google Maps API is used to display the vehicle on the map in the Smart phone application. It shows where exactly the bus is there on the map and provides updated information to the user at different time intervals. This also displays the estimated arrival time, which helps the user to know when exactly the bus is going to reach his/her stop.

Apart from this our system also provides the congestion details for driver. It provides the alternate routes for the driver if there is any congestion in his route. The next major advantage is, if the bus is in any emergency situation it sends an alert message to the transport in charge. The user can get flexibility of planning travel using the app, to decide when to catch the bus. The proposed system is user friendly and ensures safety and surveillance at low maintenance cost.

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CHEPTER 1

INTRODUTION TO DOMAIN

Vehicle tracking systems were first implemented for the shipping industry because people wanted to know where each vehicle was at any given time. These days, however, with technology growing at a fast pace, automated vehicle tracking system is being used in a variety of ways to track and display vehicle locations in real-time.

However, bus transportation service has very poor transportation information system nowadays. Bus users do not know the exact arrival time for a bus, but only know the scheduled approximate arrival time: Bus transportation service does not have a p a proper system to track all buses position and the actual arrival time in every bus stop. These problems occur because current bus service system did not apply real time tracking technology to track on each buses on the road and also lack of a platform to update latest bus traffic information to bus users.

In order to solve these problems and enhance current bus service system, real time bus tracking system has to develop and implement. With real time bus tracking system, bus position data is connected real time and transmitted to a central server for processing and extracting transit information. The main technology used to develop this system is Global Positioning System (GPS), GPS technology able to receives the position of an object from space-based satellite navigation system through a GPS receiver.

For wireless data transmission, GSM and SMS technology are commonly used. The SMS technology through GSM network and GSM modem provide a user with vehicle location information. Instead of using SMS, the bus tracking system uses the smart phone application to track and monitor a bus location obtained from the in-vehicle tracking device.

Smart Bus Tracking System Introduction Department of IT, GECG - Gandhinagar 2022-23. The bus location is automatically placed on Google maps, which makes it easier for tracking a vehicle and provides users with more accurate vehicle location information. The developed bus tracking system will be able to provide bus users a real time platform to check on updated bus traffic information, for example bus arrival or departure time. Besides, this system also reduces workload for bus management team and provides an immediate platform to update latest and accurate bus traffic information to bus users.

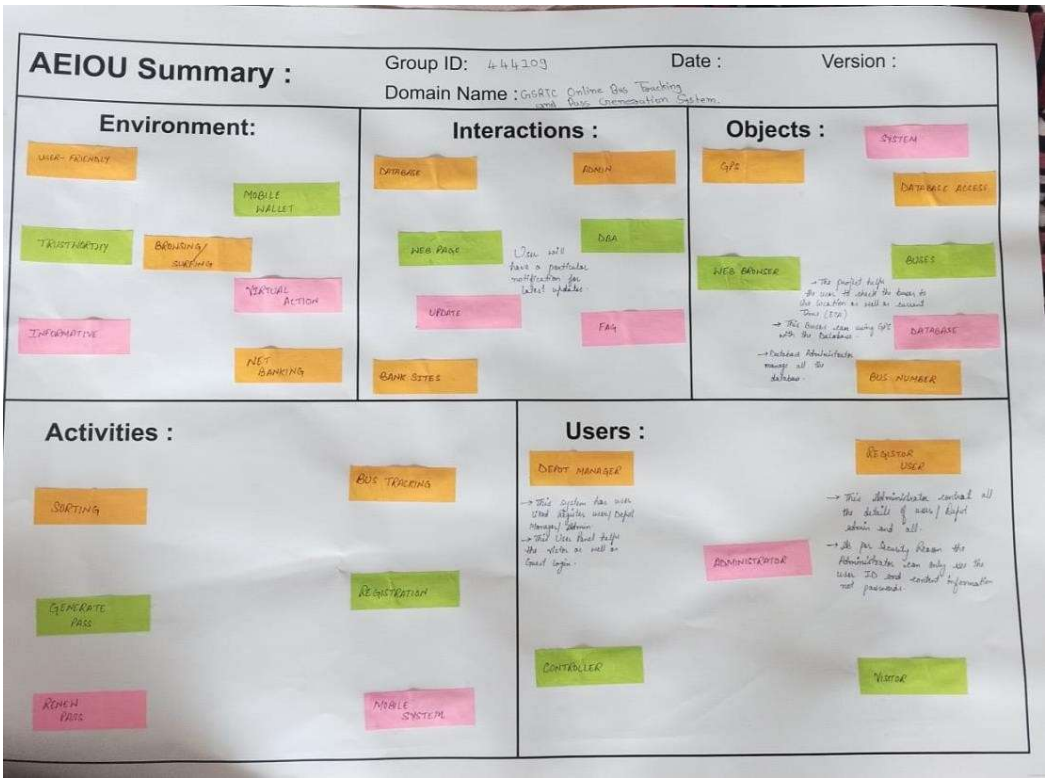
CHAPTER 2

PHASE 1

REVERSE ENGINEERING

2.1 Modification in canvases based on reverse engineering

2.1.1 AEIOU Canvas and description



<u>Summary of AEIOU Canvas and their observations.</u>

ACTIVITIES:- Bus Track, New Pass Generate, Register User, Visitor, Renew Pass

ENVIRONMENT: - User Friendly, Trustworthy,

Informative, Mobile Wallet,

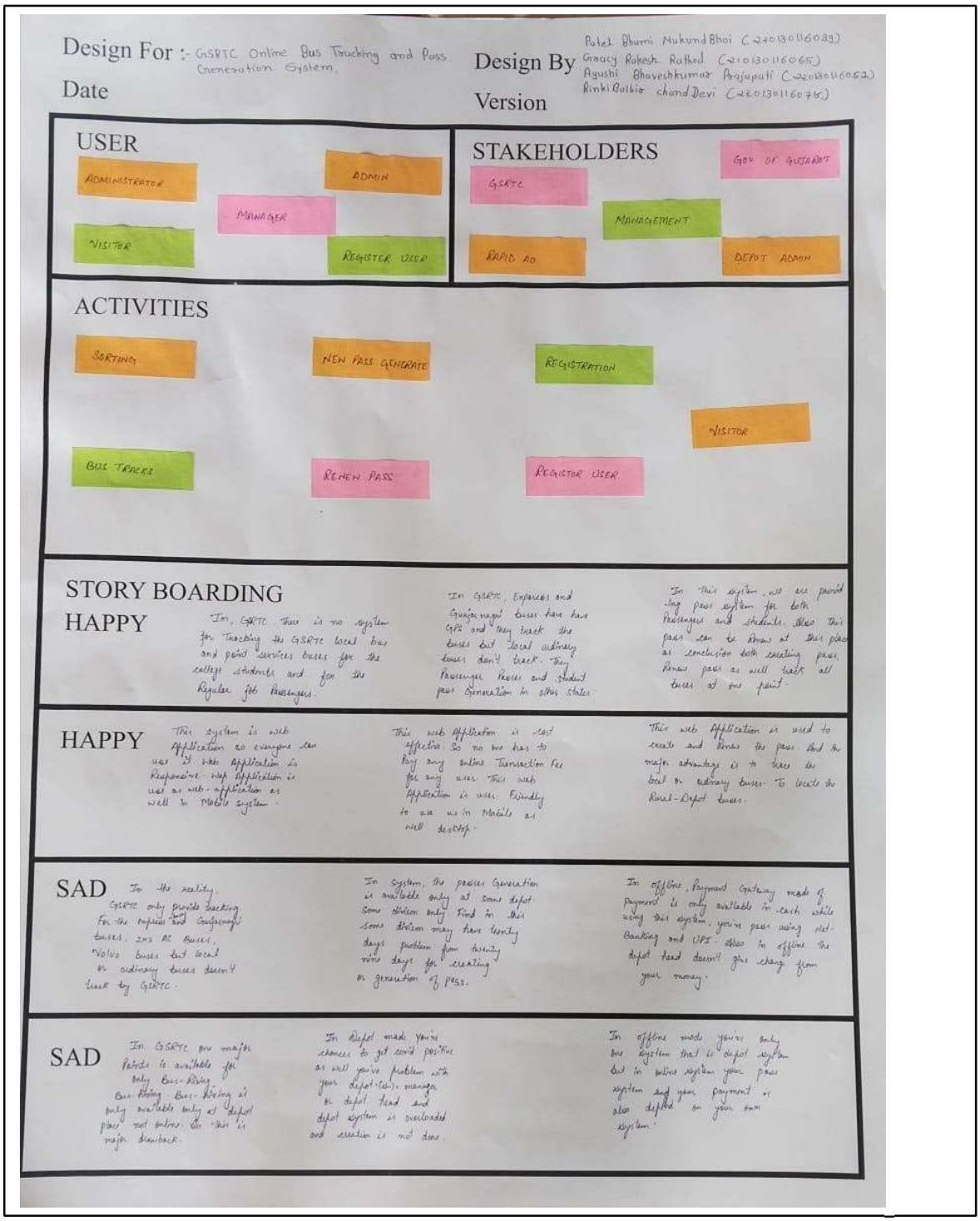
Surfing, Efficient

INTERACTIONS: Developer, Register User, Visitor or Guest User, Payment Gateway, Depot Admin, Depot Manager

OBJECTS: -GPS, Browser, Bus, Bus Number, Payment Gateway, System

USERS: - Admin, Manager, Register User, Visitor, Developer

2.2.2 Empathy canvas images and description



Summary of Empathy Canvas and their observations.

In empathy canvas, we have covered different areas like who would be the users, who are the stakeholders, what are the activities and what are the story boarding associated with our topic.

USER: -

In this we have Register users, visitor, administrator, agents, advertisers, web developer, researchers, etc.

STACKHOLDERS: -

Here we have, GSRTC Head Depot, developers, State Government. RapidGO, coders, programmers, IT security, etc.

ACTIVITIES:-

In ACTIVITIES, we have, payment options, surfing, special offers, product reviews, feedback collection, refunds, providing security, bus tracking, basic activities like create a new pass and renew it.

2.2.3 Ideation canvas image and description

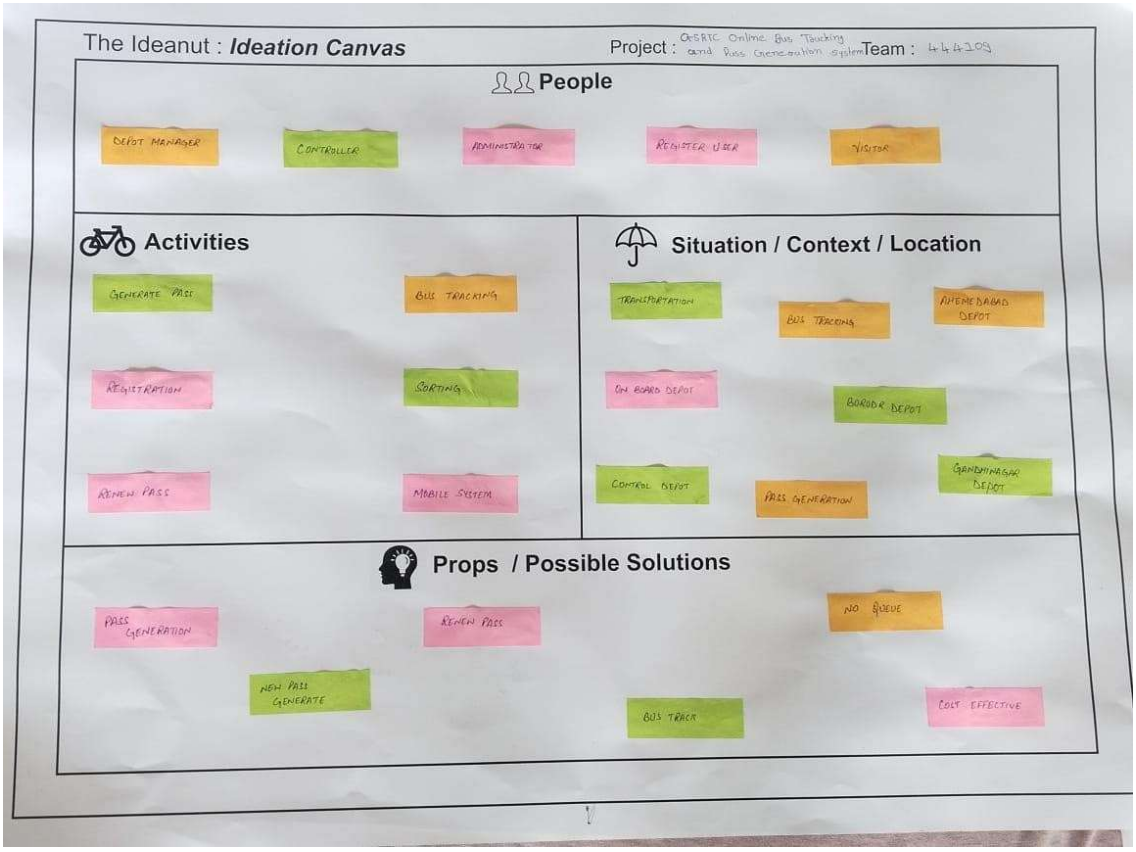


Figure 2.2.3: Ideation Canvas

Summary of Ideation Canvas and their observations.

In ideation canvas we observed about people, their activity, situation/context/location. and their possible solutions.

PEOPLE:-

People includes Register users, Visitor, administrator, web developer, depot manager, stakeholders, depot admin, transport minister, Bus staff, GSRTC staff.

ACTIVITIES:-

There are various activities in "GSRTC ONLINE BUS TRACKING AND PASS GENERATION SYSTEM

"like create a pass and renew the pass. In this activity the user would generate the desire pass and the GSRTC staff would create the pass. Other activity is payment; we provide a variety of payment options so that it provides an easy for the user in payment activity. Other activity is bus tracking, here the user can track the bus that he/she has want to go and can also keep a track of it. Other activity is security, we try to provide a very secured and trust-worthy platform so that the buyers feel it safe and secured to use

SITUATION/CONTEXT/LOCATION:-

In situation we are depicting the various situation where we can use our web. Developers and researchers test and analyses psychological effect.

In context as we create the pass and we confirm it and then we track of passes and then when it done, we pay it with our continent payment option.

In location we have GSRTC, database, web.

PROPS/POSSIBLE SOLUTIONS: -

In props/possible solutions HTML, Dot Net, image editor, internet, logo design, templet design, magazine, catalogue, etc.

2.2.4 Product development canvas image and description

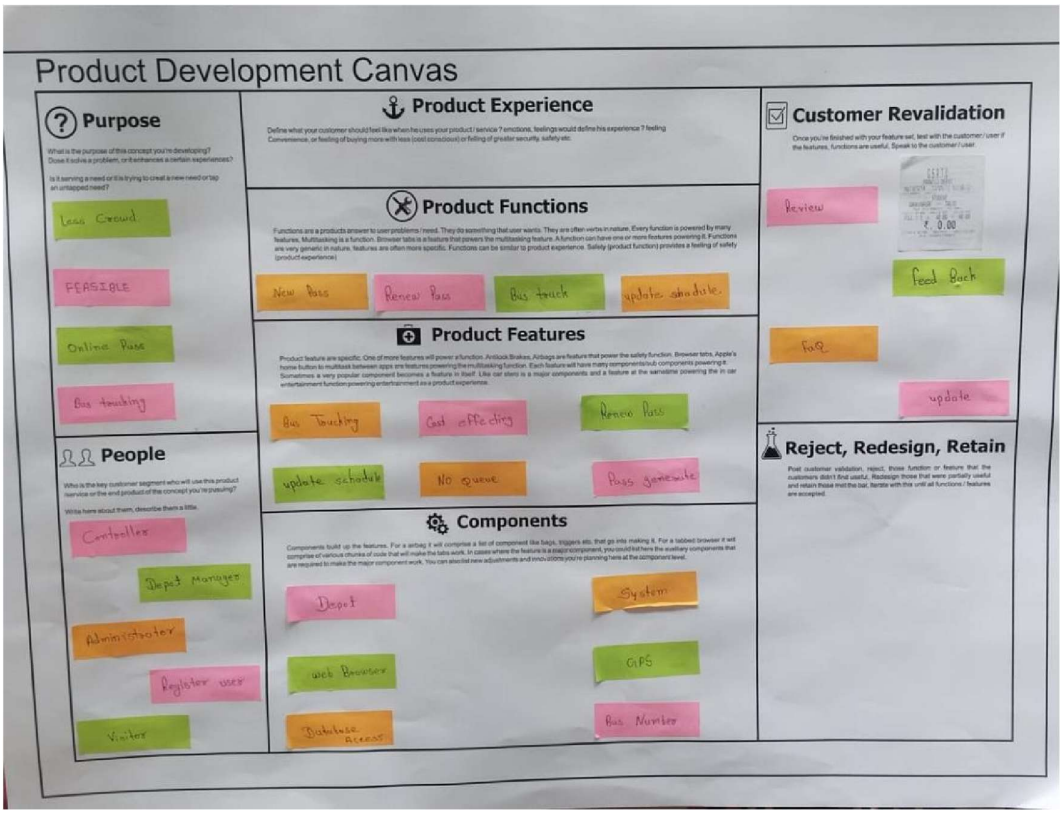
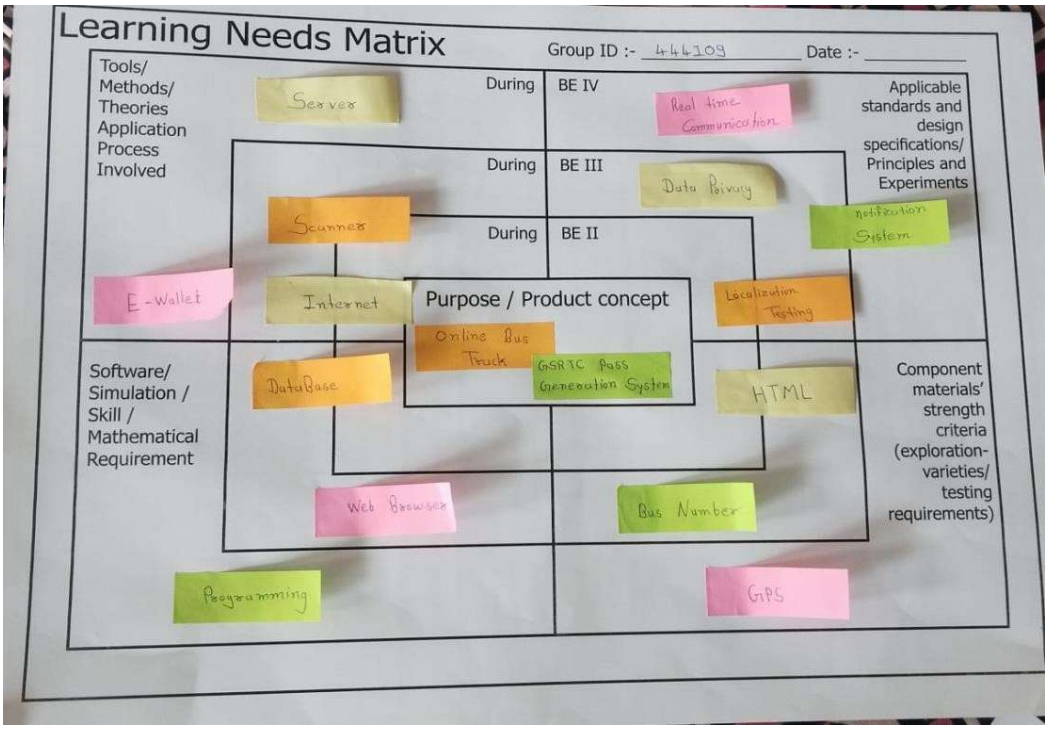


Figure 2.2.4: Product Development Canvas

2.2.5 Learning Needs Metrix



Development Canvas and their observations.

In this canvas we define the purpose of our product, people using it, experience, features, component used, customer revalidation, and whether we must reject, redesign or retain our design for our product.

PURPOSE: -

The purpose of our product is to decrease the queue at depot as well to save the time of people.

PEOPLE:-

People includes register users, visitors, depot admin, depot manager, transport minister, web designer, DBA.

PRODUCT EXPERIENCE:-

The experience by the users were innovative, provides quick services, proved to be one of the healthy browsing technique, mainly provides Indian art forms on-hand.

PRODUCT FUNCTION: -

We provide our web available in various languages so that it is easy for the user. Helps in searching different pass items. The users can track their bus. It provides a very easy platform for communication too.

PRODUCT FEATURES: -

The features of our product are it track bus, provides create pass, refund and exchange policy, and renew pass also has two factor authentications, we have made it user-friendly and user-oriented, and we have made it language flexible.

COMPONENTS:-

The components used are digital gadgets, serves, programming, catalogue, software upgrader, barcode scanner, image editor, HTML, Dot Net, website hosting, website template design, domain name.

CUSTOMER REVALIDATION: -

Introduce time tracking facility.

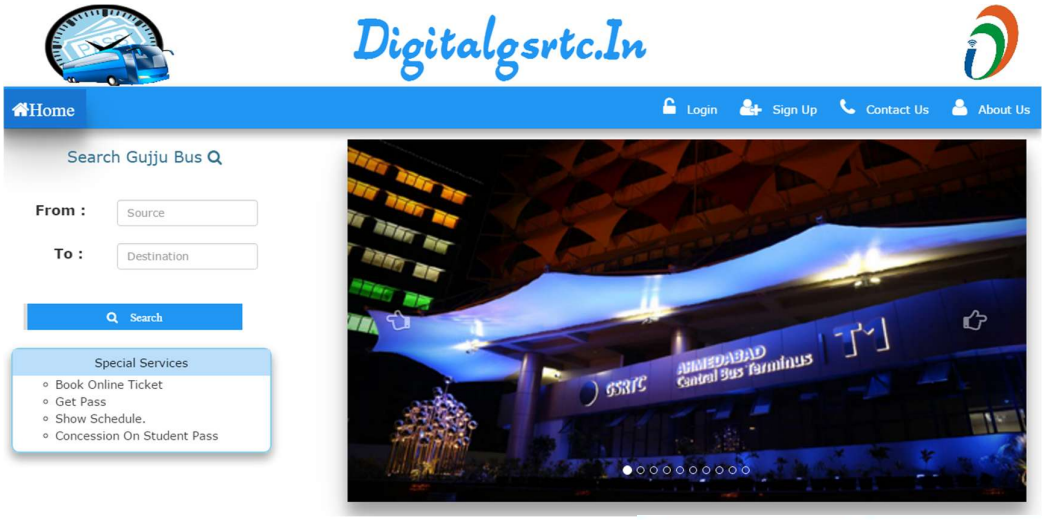
REJECT, REDESIGN, RETURN: -

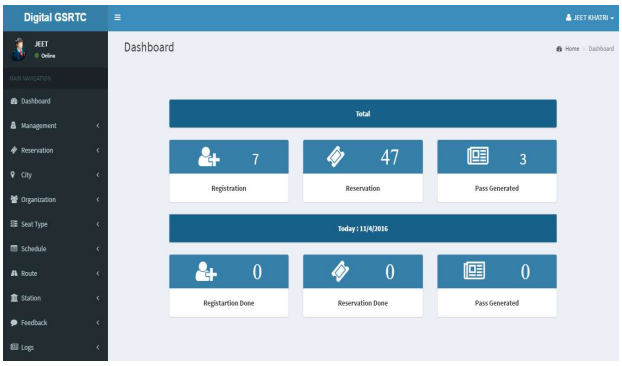
Retain

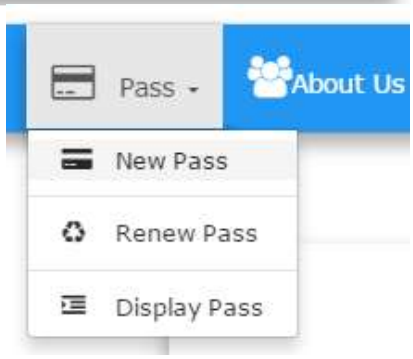
CHAPTER 3

PHASE 2 ROUGH PROTOTYPE MODEL

3.1 Prototype Image







4.1 Mind MAP

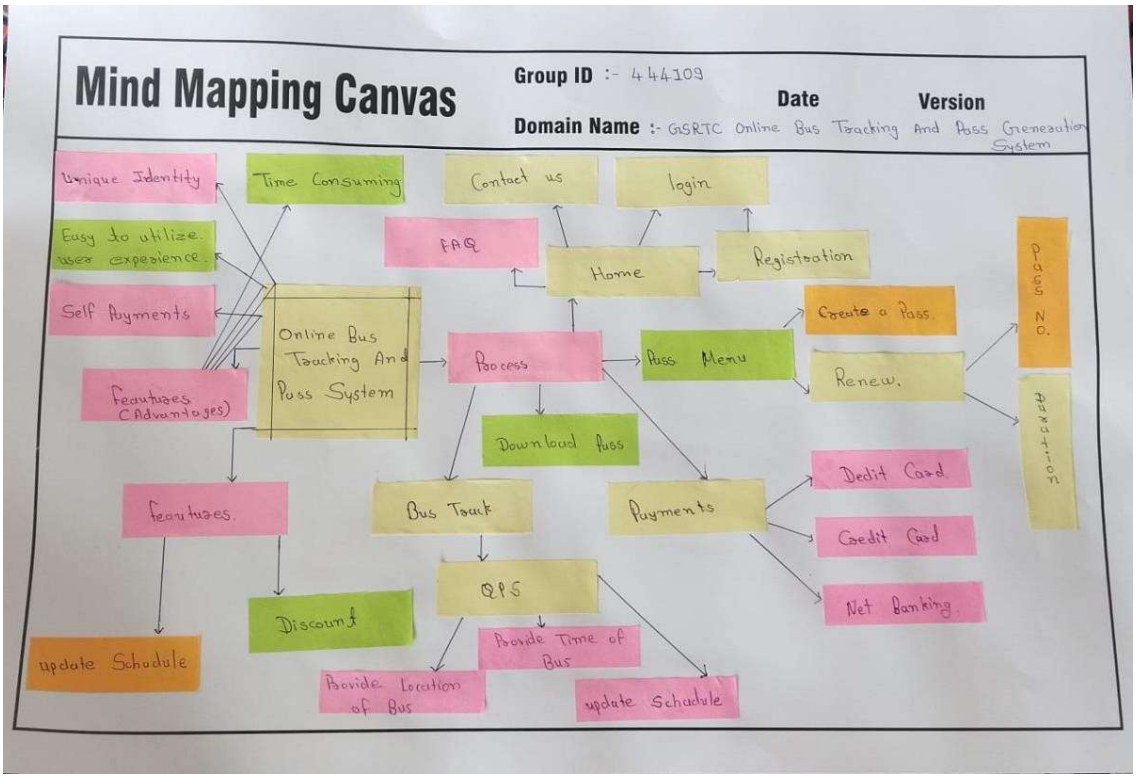


Figure 4.1 MIND MAP

References

[1] https://www.google.com/

[2] https://gsrtc.in/site/